

# Linnet

Half-edge graphs and subgraph-first algorithms

## 1 Overview

Linnet is a graph library for tensor networks, Feynman diagrams, and other workflows where the boundary of a subgraph matters. Its central representation is a half-edge graph. An ordinary internal edge is a pair of half-edges, while an external edge has one dangling half-edge. A node owns incident half-edges, and a subgraph is a selection of half-edges rather than a detached copy of the graph.

### Why half-edges

Cutting an internal edge can turn its two halves into boundary edges without changing the degree of either endpoint. This makes operations such as extraction, excision, contraction, sewing, traversal, and cut analysis natural for physics graphs. Linnet algorithms therefore accept subgraph views even when the caller wants to operate on the full graph.

### 1.1 A working model

A Linnet workflow has four parts:

- construct a graph with `HedgeGraphBuilder`, or parse a DOT graph;
- retain the typed identifiers returned for half-edges, nodes, and edges;
- describe the region of interest with one of the subgraph representations;
- run traversal or mutation operations against that subgraph and the graph's involution.

The involution is the pairing map for half-edges. A paired entry represents an internal edge; an identity entry represents a dangling edge and records its flow. Edge orientation and flow are related concepts, but they are not interchangeable. Consult the API reference for the exact variants and conversion rules in your Linnet release.

### Indexes belong to one graph

`Hedge`, `NodeIndex`, and `EdgeIndex` are compact typed indexes, not durable object IDs. Graph edits can change storage and mappings. Keep the result returned by an operation, and do not apply an index or subgraph bitset to an unrelated graph just because its numeric values happen to fit.

### 1.2 Construction and interchange

The Rust package can be added directly from crates.io:

```
[dependencies]
linnet = "0.17"
```

Programmatic construction uses `HedgeGraphBuilder`. DOT support provides a convenient human-readable interchange and debugging format; it can carry global, node, edge, and half-edge attributes. Treat DOT as a representation boundary, not as a substitute for the typed graph invariants. Validate or handle parser errors before running algorithms.

The drawing feature adds graph drawing and layout support. Layout is separate from graph identity: coordinates and rendering attributes may change without changing the topology.

## 1.3 Related crates

### One graph layer, several consumers

Spensio uses Linnet for tensor-network connectivity and adds tensor contraction semantics. GammaLoop uses the graph model in collider workflows and adds process generation, integration, and persistent state.

The Linnet [README](#) provides a compact crate summary; the Rust API reference covers individual types and methods.

## 2 Tutorial

In this tutorial, you will construct a half-edge graph in Rust, validate its invariants, find its one independent cycle, and print DOT. The builder returns Linnet's typed node and half-edge identifiers directly, so you can see where graph identity enters the workflow.

### 2.1 Prerequisites

Create a Rust binary project and add the current Linnet release:

```
cargo new linnet-first-graph
cd linnet-first-graph
cargo add linnet@0.17
```

No drawing feature or external layout program is needed to construct the graph and emit DOT.

### 2.2 Build a graph with a boundary

Replace `src/main.rs` with the following program:

```
use linnet::half_edge::{
    builder::HedgeGraphBuilder,
    involution::Flow,
    HedgeGraph,
};

fn main() {
    let mut builder = HedgeGraphBuilder::new();
    let a = builder.add_node("a");
    let b = builder.add_node("b");
    let c = builder.add_node("c");

    builder.add_edge(a, b, "ab", false);
    builder.add_edge(b, c, "bc", false);
    builder.add_edge(c, a, "ca", false);
    builder.add_external_edge(a, "incoming", false, Flow::Sink);
    builder.add_external_edge(c, "outgoing", false, Flow::Source);

    let graph: HedgeGraph<&str, &str> = builder.build();
    graph.check().expect("the half-edge involution is valid");

    let (basis, _) = graph.cycle_basis();
    assert_eq!(basis.len(), 1);
    println!("{}", graph.base_dot());
}
```

Run it with `cargo run`. Success means the assertion passes and `stdout` contains a DOT digraph with nodes `a`, `b`, and `c`, three paired internal edges, and two dangling boundary edges. The external edges do not add a cycle, so the triangle's cycle-basis rank remains one.

#### Keep the returned indexes with their graph

`a`, `b`, and `c` are compact `NodeIndex` values owned by this graph construction. Do not serialize their numeric representation as a durable identity or reuse them against another graph. Mutating operations return the mappings needed to relate old and new storage.

## 2.3 Make the first useful variation

Add a fourth internal edge between `a` and `b`, run the program again, and change the assertion to `basis.len() == 2`. This demonstrates the key distinction: parallel paired half-edges add an independent cycle, while dangling identity half-edges describe the graph boundary.

From here, use a subgraph view to run traversal, cuts, or contraction against only the region you intend to modify. Enable Linnet's drawing feature only when you need layout rather than plain DOT interchange.

## 2.4 Troubleshooting and next steps

- A type-inference error around `build()` means Rust cannot choose node storage or data types; keep the explicit `HedgeGraph<&str, &str>` annotation from the example.
- A failed `check()` points to an involution or storage invariant. Validate immediately after parsing or construction, before applying graph algorithms.
- If a traversal includes an unexpected dangling edge, check the selected subgraph and the `Flow` assigned to each external edge.
- Continue with the subgraph-and-algorithms manual, then consult the Rust API reference for the exact return mappings of mutating operations and additional builder methods.

# 3 Subgraphs, traversal, cuts, and drawing

Linnet algorithms operate on a graph together with an explicit subgraph view. This lets one algorithm work on the full graph, a selected region, or the boundary created by cutting paired half-edges. Choose the subgraph representation by operation: bit-backed sets are efficient for repeated set algebra, while borrowed or mapped views avoid copying when identity must remain attached to the owning graph.

## 3.1 Traversal and connectivity

A traversal needs a starting half-edge or node, the active subgraph, and a policy describing which adjacent half-edges may be crossed. Results retain typed indexes into the original graph. Breadth-first traversal is useful for distance layers and connectivity; depth-first traversal is useful for trees, back edges, and decompositions. A traversal tree is a derived view, not a new graph, so mutations must be coordinated with the indexes it contains.

Connected components, bridges, articulation structure, and biconnected regions all depend on the chosen subgraph. Run them on the full graph only when external half-edges and excluded nodes should participate in the answer.

## 3.2 Cycles and cuts

`cycle_basis` returns an independent basis and rank information. `all_cycles` expands that basis and can grow exponentially; prefer a basis when assigning loop momenta or computing a loop count. Symmetric differences combine cycle bitsets without inventing new graph identities.

Cut enumeration separates required left and right node sets and returns the left region, cut content, and right region. A cut edge is represented through its half-edges, preserving which side owns each endpoint. Validate

direction/flow after enumeration when a physical Cutkosky or tensor-network interpretation depends on orientation.

#### **Complexity is part of the API choice**

Enumerating every cycle or every admissible cut is inherently combinatorial. Filter the subgraph and endpoint sets first, and use basis or connectivity operations when enumeration is not required by the caller.

### **3.3 Mutation boundaries**

Extraction copies a selected region, excision separates a graph along its boundary, sewing joins compatible dangling half-edges, and contraction identifies structure. These operations return mappings or replacement indexes where needed. Keep those results: a numeric Hedge or NodeIndex from before a storage-changing operation is not automatically a valid index into the resulting graph.

### **3.4 DOT and drawing**

DOT parsing and emission are interchange/debugging boundaries. Attributes can carry graph, node, edge, and half-edge data, but the typed Rust invariants are reconstructed by the parser; handle parser errors before running algorithms. The drawing feature adds layout and rendering. Layout coordinates are presentation data and may vary without changing topology, subgraph membership, or edge flow.

#### **Tensor contraction uses Spenso**

Linnet determines connectivity and graph transformations. Tensor compatibility, contraction cost, and execution belong to Spenso even when a Spenso network delegates its graph algorithms to Linnet.

## **4 Rust and Python APIs**

### **4.1 Rust package**

The `linnet` package is organized into these principal areas:

- `half_edge` provides HedgeGraph, builders, the involution and edge data, subgraph types, traversal trees, and graph algorithms;
- `parser` provides DOT parsing and DOT-backed graph data;
- `drawing` provides rendering data and layout helpers;
- `permutation`, `tree`, and `union_find` provide supporting algorithms and stores.

The generic graph types separate edge, vertex, and half-edge payloads from the node-storage implementation. Start with the [curated Rust API](#), then follow its Rustdoc links for complete generic bounds and method signatures; choose a concrete storage type according to the workflow and mutation behavior you need.

Cargo features enable optional capabilities:

- `serde` adds serialization traits;
- `bincode` adds binary encoding support;
- `drawing` adds the drawing stack;
- `symbolica` adds Symbolica interoperability.

An item shown in an all-features API reference may not be available in a default build. Check its required feature and your `Cargo.toml` before using it.

## 4.2 Standalone Python distribution

### Distribution and import have different names

Install the Python distribution `linnet-py`, then import `linnet_py`. It requires Python 3.10 or newer. It is a standalone extension and is not a `symbolica.community` module. Its package version is independent of the Rust `linnet` version. Record both versions when diagnosing compatibility between Rust and Python code.

The Python binding focuses on DOT-backed graphs and mirrors typed identifiers and subgraphs:

```
from linnet_py import DotGraphBuilder

builder = DotGraphBuilder()
left = builder.add_node()
right = builder.add_node()
builder.add_edge(left, right)
graph = builder.build()

print(graph.dot())
```

`DotGraph` can also parse one graph from a string or file, or a set of graphs from a string. The exported classes include `Hedge`, `NodeIndex`, `EdgeIndex`, `Flow`, `Orientation`, graph attribute wrappers, `HedgePair`, `Subgraph`, `Cycle`, `OrientedCut`, `TraversalTree`, `DotGraph`, and `DotGraphBuilder`.

Mutable statement wrappers expose mapping-like access to DOT attributes. Consult the structured [Python API](#) for their lifetime and write-through behavior, and do not assume that a copied mapping remains attached to its graph.

## 5 Releases and change history

The `Linnet` changelog records changes to the Rust crate. The standalone `linnet-py` distribution has its own version and does not currently have a separate changelog in this repository.

The changelog records user-visible graph, parser, drawing, and algorithm changes. In particular, users moving across releases should check index ordering, DOT serialization, subgraph behavior, and feature-gated integration changes rather than assuming a serialized graph or incidental ordering is stable.

### 5.1 Choosing the matching documentation

The latest channel may describe unreleased behavior. Documentation under `snapshots/<tag>` is fixed to a tagged repository revision. Choose a snapshot whose component information lists your Rust crate version, and check the `linnet-py` version separately when using the Python bindings.

#### Before upgrading

Review changes to index ordering, DOT serialization, subgraph behavior, and enabled features. Re-validate serialized graphs and any code that depends on traversal or iteration order.

## API reference

The packages available in this version are listed below. Follow an item link for its complete language reference.

### linnet

Linnet's supported half-edge graph API. Algorithms are grouped in the order used by the manual and full Rustdoc.

#### Half-edge graph

Stores graph topology as paired half-edges with independent edge, vertex, and half-edge payloads.

The main graph data structure, representing a graph using the half-edge (or doubly connected edge list - DCEL) principle.

A `HedgeGraph` stores nodes and edges (represented as pairs of half-edges). It is generic over the data associated with edges (`E`), nodes/vertices (`V`), and the specific storage strategy used for nodes (`S`).

The half-edge representation allows for efficient traversal of graph topology, such as iterating around faces (in planar embeddings), finding incident edges to a node, or quickly accessing an edge's opposite half-edge.

#### # Type Parameters

- `E`: The type of data associated with each edge (or pair of half-edges).
- `V`: The type of data associated with each node (vertex).
- `S`: The node storage strategy, implementing the `NodeStorage` trait.

This determines how node data and their connectivity to half-edges are stored. Defaults to `DefaultNodeStore<V>` (feature-selected; `nodestore-vec` uses `nodestore::NodeStorageVec<V>`).

```
#[derive(Clone, Debug, PartialEq, Eq, PartialOrd, Ord, Hash)]
#[cfg_attr(feature = "serde", derive(serde::Serialize, serde::Deserialize))]
#[cfg_attr(feature = "bincode",
derive(bincode_trait_derive::Encode, bincode_trait_derive::Decode))]
#[cfg_attr(feature = "rkyv",
derive(rkyv::Archive, rkyv::Serialize, rkyv::Deserialize))]
#[cfg_attr(feature = "rkyv", archive(check_bytes))]
#[doc =
" The main graph data structure, representing a graph using the half-edge"
#[doc = " (or doubly connected edge list - DCEL) principle." ] #[doc = ""]
#[doc =
" A HedgeGraph stores nodes and edges (represented as pairs of half-edges)."]
#[doc =
" It is generic over the data associated with edges (E), nodes/vertices (V),"]
#[doc = " and the specific storage strategy used for nodes (S)."]
#[doc = ""]
#[doc =
" The half-edge representation allows for efficient traversal of graph topology,"
#[doc =
" such as iterating around faces (in planar embeddings), finding incident edges"
#[doc = " to a node, or quickly accessing an edge's opposite half-edge." ]
#[doc = ""] #[doc = " # Type Parameters"] #[doc = ""]
#[doc =
" - E: The type of data associated with each edge (or pair of half-edges)."]
```

```

#[doc = " - `V`: The type of data associated with each node (vertex)."]
#[doc =
" - `S`: The node storage strategy, implementing the [`NodeStorage`] trait."]
#[doc =
" This determines how node data and their connectivity to half-edges"]
#[doc =
" are stored. Defaults to [`DefaultNodeStore<V>`] (feature-selected; `nodestore-vec` uses")
#[doc = "    [`nodestore::NodeStorageVec<V>`]]."] pub struct HedgeGraph < E, V,
H = NoData, S : NodeStorage < NodeData = V > = DefaultNodeStore < V > >
{
    pub(crate) hedge_data : HedgeVec < H > ,
    #[doc =
" Internal storage for all half-edges, their data, and their topological"]
    #[doc =
" relationships (e.g., opposite half-edge, next half-edge around a node)."]
    #[doc = " This is typically a [`SmartHedgeVec<E>`]]."] pub(crate)
    edge_store : SmartEdgeVec < E > ,
    #[doc =
" Storage for all nodes in the graph, including their data (`V`) and"]
    #[doc = " information about the half-edges incident to them."]
    #[doc =
" The specific implementation is determined by the `S` type parameter."]
    pub node_store : S,
}

```

**Reference feature matrix:** nodestore-vec, serde, bincode, drawing, symbolica (item is available without an additional gate)

### Supported usage

```
use linnet::half_edge::HedgeGraph;
```

#### **hedge\_data**

**Kind:** Field

HedgeVec < H >

#### **edge\_store**

**Kind:** Field

SmartEdgeVec < E >

Internal storage for all half-edges, their data, and their topological relationships (e.g., opposite half-edge, next half-edge around a node). This is typically a [`SmartHedgeVec<E>`].

#### **node\_store**

**Kind:** Field

S

Storage for all nodes in the graph, including their data (`V`) and information about the half-edges incident to them. The specific implementation is determined by the `S` type parameter.

Exhaustive reference · Source

## Graph builder

Builds a valid half-edge graph while assigning typed node, edge, and hedge identifiers.

A builder for programmatically constructing [`HedgeGraph``] instances.

This builder allows for the incremental addition of nodes and edges (both paired and external/dangling) before finalizing the graph structure.

### # Type Parameters

- ``E``: The type of data to be associated with edges.
- ``V``: The type of data to be associated with nodes.

### # Example

```
```rust,ignore
use linnet::half_edge::builder::HedgeGraphBuilder;
use linnet::half_edge::involution::{Flow, Orientation};

let mut builder = HedgeGraphBuilder::<&str, &str>::new();

let node0 = builder.add_node("Node_0_Data");
let node1 = builder.add_node("Node_1_Data");
let node2 = builder.add_node("Node_2_Data");

builder.add_edge(node0, node1, "Edge_01_Data", Orientation::Default);
builder.add_external_edge(node2, "External_Edge_Data", Orientation::Undirected,
Flow::Source);

// Assuming a NodeStorage type MyNodeStore is defined and implements NodeStorageOps
// let graph: HedgeGraph<&str, &str, MyNodeStore> = builder.build();
```

#[derive(Clone, Debug)]
#[cfg_attr(feature = "rkyv",
derive(rkyv::Archive, rkyv::Serialize, rkyv::Deserialize))]
#[doc =
" A builder for programmatically constructing [HedgeGraph`] instances."]
#[doc = ""]
#[doc =
" This builder allows for the incremental addition of nodes and edges (both"
#[doc =
" paired and external/dangling) before finalizing the graph structure."
#[doc = ""] #[doc = " # Type Parameters"] #[doc = ""]
#[doc = " - `E`: The type of data to be associated with edges."
#[doc = " - `V`: The type of data to be associated with nodes."] #[doc = ""]
#[doc = " # Example"] #[doc = ""] #[doc = " ```rust,ignore"]
#[doc = " use linnet::half_edge::builder::HedgeGraphBuilder;"]
#[doc = " use linnet::half_edge::involution::{Flow, Orientation};"]
#[doc = ""]
#[doc = " let mut builder = HedgeGraphBuilder::<&str, &str>::new();"]
#[doc = ""] #[doc = " let node0 = builder.add_node(\"Node_0_Data\");"]
#[doc = " let node1 = builder.add_node(\"Node_1_Data\");"]
#[doc = " let node2 = builder.add_node(\"Node_2_Data\");"] #[doc = ""]
#[doc =
" builder.add_edge(node0, node1, \"Edge_01_Data\", Orientation::Default);"]
```



```

#[doc =
" builder.add_external_edge(node2, \"External_Edge_Data\", Orientation::Undirected,
Flow::Source);"]
#[doc = ""]
#[doc =
" // Assuming a NodeStorage type MyNodeStore is defined and implements
NodeStorageOps"]
#[doc =
" // let graph: HedgeGraph<&str, &str, MyNodeStore> = builder.build();"]
#[doc = " ````"] pub struct HedgeGraphBuilder < E, V, H = NoData >
{
    hedge_data : HedgeVec < H > ,
    #[doc =
    " A list of nodes currently being built, stored as [`HedgeNodeBuilder`]
instances."]
    nodes : Vec < HedgeNodeBuilder < V > > ,
    #[doc =
    " The [`Involution`] structure managing the half-edges being added to the
graph."]
    pub(crate) involution : Involution < E > ,
}

```

**Reference feature matrix:** nodestore-vec, serde, bincode, drawing, symbolica (item is available without an additional gate)

### Example 1

```

use linnet::half_edge::builder::HedgeGraphBuilder;
use linnet::half_edge::involution::{Flow, Orientation};

let mut builder = HedgeGraphBuilder::<&str, &str>::new();

let node0 = builder.add_node("Node_0_Data");
let node1 = builder.add_node("Node_1_Data");
let node2 = builder.add_node("Node_2_Data");

builder.add_edge(node0, node1, "Edge_01_Data", Orientation::Default);
builder.add_external_edge(node2, "External_Edge_Data", Orientation::Undirected,
Flow::Source);

// Assuming a NodeStorage type MyNodeStore is defined and implements NodeStorageOps
// let graph: HedgeGraph<&str, &str, MyNodeStore> = builder.build();

```

### Supported usage

```
use linnet::half_edge::builder::HedgeGraphBuilder;
```

#### **hedge\_data**

**Kind:** Field

HedgeVec < H >

#### **nodes**

**Kind:** Field

Vec < HedgeNodeBuilder < V > >

A list of nodes currently being built, stored as [`HedgeNodeBuilder`] instances.

## involution

**Kind:** Field

Involution < E >

The [``Involution``] structure managing the half-edges being added to the graph.

Exhaustive reference · Source

## Bit-vector subgraph

Represents a subgraph as an efficient half-edge inclusion filter.

Represents a subgraph as an efficient half-edge inclusion filter.

```
pub type SuBitGraph = SubSet < Hedge > ;
```

**Reference feature matrix:** `nodestore-vec`, `serde`, `bincode`, `drawing`, `symbolica` (item is available without an additional gate)

## Supported usage

```
use linnet::half_edge::subgraph::SuBitGraph;
```

Exhaustive reference · Source

## Traversal tree

Records parent, root, and traversal information produced by breadth-first or depth-first graph walks.

Represents a traversal tree (e.g., DFS or BFS tree) derived from a [``HedgeGraph``].

This structure uses a generic [``Forest``] to store the tree topology. Each "root" in the

``Forest`` corresponds to a node in the original ``HedgeGraph``, and its data is a [``TTRoot``]

enum indicating its role (Root, Child, or None) in this specific traversal. The nodes within each tree of the ``Forest`` typically represent the half-edges that form the traversal path.

## # Type Parameters

- ``P``: The type of [``ForestNodeStore``] used by the underlying ``Forest``.

It stores ``()`` as ``NodeData`` because the tree nodes (representing half-edges) don't need additional data beyond their structural role in the traversal tree.

Defaults to [``ParentPointerStore<()>``].

```
#[derive(Debug, Clone)]
```

```
#[cfg_attr(feature = "serde", derive(serde::Serialize, serde::Deserialize))]
```

```
#[doc =
```

```
" Represents a traversal tree (e.g., DFS or BFS tree) derived from a  
[`HedgeGraph`]."]
```

```
#[doc = ""]
```

```
#[doc =
```

```
" This structure uses a generic [`Forest`] to store the tree topology. Each \"root\"  
in the"]
```

```
#[doc =
```

```
" `Forest` corresponds to a node in the original `HedgeGraph`, and its data is a  
[`TTRoot`]"
```

```
#[doc =
```

```
" enum indicating its role (Root, Child, or None) in this specific traversal. The  
nodes"]
```

```

#[doc =
" within each tree of the `Forest` typically represent the half-edges that form the"
#[doc = " traversal path."] #[doc = ""] #[doc = " # Type Parameters"]
#[doc = ""]
#[doc =
" - `P`: The type of [`ForestNodeStore`] used by the underlying `Forest`."]
#[doc =
" It stores `()` as `NodeData` because the tree nodes (representing half-edges)"
#[doc =
" don't need additional data beyond their structural role in the traversal tree."]
#[doc = " Defaults to [`ParentPointerStore<()>`]."] pub struct
SimpleTraversalTree < P : ForestNodeStore < NodeData = () > =
ParentPointerStore < () > >
{
    #[doc =
    " The underlying forest structure representing the traversal tree(s)."]
    #[doc =
    " Each root in this forest is a graph node, and its data is its [`TTRoot`]
status."]
    #[doc = " Nodes within each tree of this forest represent half-edges."]
    forest : Forest < TTRoot, P > ,
    #[doc =
    " A bitmask representing the set of half-edges that constitute the actual"
    #[doc = " edges of this traversal tree."] pub tree_subgraph : SuBitGraph,
}

```

**Reference feature matrix:** nodestore-vec, serde, bincode, drawing, symbolica (item is available without an additional gate)

### Supported usage

```
use linnet::half_edge::tree::SimpleTraversalTree;
```

#### forest

**Kind:** Field

Forest < TTRoot, P >

The underlying forest structure representing the traversal tree(s).  
Each root in this forest is a graph node, and its data is its [`TTRoot`] status.  
Nodes within each tree of this forest represent half-edges.

#### tree\_subgraph

**Kind:** Field

SuBitGraph

A bitmask representing the set of half-edges that constitute the actual  
edges of this traversal tree.

Exhaustive reference · Source

#### DOT-backed graph

Combines a half-edge graph with DOT attributes and parser/serializer support.  
Combines a half-edge graph with DOT attributes and parser/serializer support.

```

#[derive(Debug, Clone, PartialEq, Eq)]
#[cfg_attr(feature = "rkyv",
derive(rkyv::Archive, rkyv::Serialize, rkyv::Deserialize))]
#[cfg_attr(feature = "rkyv", archive(check_bytes))] pub struct DotGraph < N :

```

```

NodeStorage < NodeData = DotVertexData > = DefaultNodeStore < DotVertexData >
>
{
    pub global_data : GlobalData, pub graph : HedgeGraph < DotEdgeData,
    DotVertexData, DotHedgeData, N > ,
}

```

**Reference feature matrix:** nodestore-vec, serde, bincode, drawing, symbolica (item is available without an additional gate)

### Supported usage

```
use linnet::parser::DotGraph;
```

#### global\_data

**Kind:** Field

GlobalData

#### graph

**Kind:** Field

HedgeGraph < DotEdgeData, DotVertexData, DotHedgeData, N >

Exhaustive reference · Source

### linnet-py

Exports registered by linnet-py.

#### Cycle

Cycle represented as a subgraph and optional loop count.

Cycle represented as a subgraph and optional loop count.

```
class Cycle:
```

**Requires features:** extension-module, abi3-py310

#### Inspect the registered export

```
from linnet_py import Cycle
help(Cycle)
```

#### filter

**Kind:** Getter

```
def filter(self) -> Subgraph:
```

Subgraph filter for this cycle.

#### loop\_count

**Kind:** Getter

```
def loop_count(self) -> typing.Optional[builtins.int]:
```

Optional loop count for this cycle.

#### \_\_repr\_\_

**Kind:** Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

Exhaustive reference · Source

### **DotEdgeData**

DOT edge data (attributes + optional edge id).

DOT edge data (attributes + optional edge id).

```
class DotEdgeData:
```

**Requires features:** extension-module, abi3-py310

### **Inspect the registered export**

```
from linnet_py import DotEdgeData
help(DotEdgeData)
```

#### **statements**

**Kind:** Getter

```
def statements(self) -> EdgeStatements:
```

Edge attributes as a dict-like proxy.

#### **local\_statements**

**Kind:** Getter

```
def local_statements(self) -> EdgeStatements:
```

Local edge attributes as a dict-like proxy.

#### **edge\_id**

**Kind:** Getter

```
def edge_id(self) -> typing.Optional[EdgeIndex]:
```

Optional edge id.

#### **\_\_new\_\_**

**Kind:** AssociatedFunction

```
def __new__(cls, statements: typing.Optional[typing.Mapping[builtins.str,
builtins.str]] = None, local_statements: typing.Optional[typing.Mapping[builtins.str,
builtins.str]] = None, edge_id: typing.Optional[builtins.int] = None) -> DotEdgeData:
```

Create edge data from fields.

#### **statements**

**Kind:** Parameter

```
statements: typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None
```

**Default:** None

#### **local\_statements**

**Kind:** Parameter

```
local_statements: typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None
```

**Default:** None

#### **edge\_id**

**Kind:** Parameter

```
edge_id: typing.Optional[builtins.int] = None
```

**Default:** None

**add\_statement**

**Kind:** Method

```
def add_statement(self, key: builtins.str, value: builtins.str) -> None:
```

Insert or update an attribute statement.

**key**

**Kind:** Parameter

key: builtins.str

**value**

**Kind:** Parameter

value: builtins.str

**\_\_repr\_\_**

**Kind:** Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

**DotGraph**

DOT-backed hedge graph.

DOT-backed hedge graph.

```
class DotGraph:
```

**Requires features:** extension-module, abi3-py310

**Inspect the registered export**

```
from linnet_py import DotGraph
help(DotGraph)
```

**global\_data**

**Kind:** Getter

```
def global_data(self) -> GlobalData:
```

Global graph attributes.

**global\_data**

**Kind:** Setter

```
def global_data(self, value: GlobalData) -> None:
```

**value**

**Kind:** Parameter

value: GlobalData

**from\_string**

**Kind:** AssociatedFunction

```
def from_string(cls, s: builtins.str) -> DotGraph:
```

Parse a DOT string into a graph.

**s**

**Kind:** Parameter

s: builtins.str

**from\_string\_set**

**Kind:** AssociatedFunction

def from\_string\_set(cls, s: builtins.str) -> builtins.list[DotGraph]:

Parse a DOT string into multiple graphs.

**s**

**Kind:** Parameter

s: builtins.str

**from\_file**

**Kind:** AssociatedFunction

def from\_file(cls, path: builtins.str) -> DotGraph:

Parse a DOT file into a graph.

**path**

**Kind:** Parameter

path: builtins.str

**debug\_dot**

**Kind:** Method

def debug\_dot(self) -> builtins.str:

Serialize graph to DOT for debugging.

**dot**

**Kind:** Method

def dot(self) -> builtins.str:

Serialize the full graph to DOT.

**dot\_of**

**Kind:** Method

def dot\_of(self, subgraph: Subgraph) -> builtins.str:

Serialize a subgraph to DOT.

**subgraph**

**Kind:** Parameter

subgraph: Subgraph

**n\_nodes**

**Kind:** Method

def n\_nodes(self) -> builtins.int:

Number of nodes.

**n\_edges****Kind:** Method

```
def n_edges(self) -> builtins.int:  
    Number of edges.
```

**n\_hedges****Kind:** Method

```
def n_hedges(self) -> builtins.int:  
    Number of hedges.
```

**n externals****Kind:** Method

```
def n_externals(self) -> builtins.int:  
    Number of external hedges.
```

**n\_internals****Kind:** Method

```
def n_internals(self) -> builtins.int:  
    Number of internal hedges.
```

**full\_filter****Kind:** Method

```
def full_filter(self) -> Subgraph:  
    Subgraph including all hedges.
```

**empty\_subgraph****Kind:** Method

```
def empty_subgraph(self) -> Subgraph:  
    Empty subgraph of this graph.
```

**iter\_edges****Kind:** Method

```
def iter_edges(self) -> builtins.list[tuple[HedgePair, EdgeIndex, EdgeData]]:  
    Iterate edges in the full graph.
```

**iter\_edges\_of****Kind:** Method

```
def iter_edges_of(self, subgraph: Subgraph) -> builtins.list[tuple[HedgePair,  
    EdgeIndex, EdgeData]]:  
    Iterate edges within a subgraph.
```

**subgraph****Kind:** Parameter

subgraph: Subgraph

**iter\_nodes****Kind:** Method



```
def iter_nodes(self) -> builtins.list[tuple[NodeIndex, builtins.list[Hedge], DotVertexData]]:
```

Iterate nodes in the full graph.

### **iter\_nodes\_of**

**Kind:** Method

```
def iter_nodes_of(self, subgraph: Subgraph) -> builtins.list[tuple[NodeIndex, builtins.list[Hedge], DotVertexData]]:
```

Iterate nodes within a subgraph.

### **subgraph**

**Kind:** Parameter

subgraph: Subgraph

### **connected\_components**

**Kind:** Method

```
def connected_components(self, subgraph: Subgraph) -> builtins.list[Subgraph]:
```

Connected components of a subgraph.

### **subgraph**

**Kind:** Parameter

subgraph: Subgraph

### **count\_connected\_components**

**Kind:** Method

```
def count_connected_components(self, subgraph: Subgraph) -> builtins.int:
```

Count connected components.

### **subgraph**

**Kind:** Parameter

subgraph: Subgraph

### **is\_connected**

**Kind:** Method

```
def is_connected(self, subgraph: Subgraph) -> builtins.bool:
```

Whether a subgraph is connected.

### **subgraph**

**Kind:** Parameter

subgraph: Subgraph

### **depth\_first\_traverse**

**Kind:** Method

```
def depth_first_traverse(self, subgraph: Subgraph, root_node: NodeIndex, include_hedge: typing.Optional[Hedge]) -> TraversalTree:
```

Depth-first traversal from a root node.

**subgraph****Kind:** Parameter

subgraph: Subgraph

**root\_node****Kind:** Parameter

root\_node: NodeIndex

**include\_hedge****Kind:** Parameter

include\_hedge: typing.Optional[Hedge]

**breadth\_first\_traverse****Kind:** Method

```
def breadth_first_traverse(self, subgraph: Subgraph, root_node: NodeIndex,
include_hedge: typing.Optional[Hedge]) -> TraversalTree:
```

Breadth-first traversal from a root node.

**subgraph****Kind:** Parameter

subgraph: Subgraph

**root\_node****Kind:** Parameter

root\_node: NodeIndex

**include\_hedge****Kind:** Parameter

include\_hedge: typing.Optional[Hedge]

**bridges****Kind:** Method

```
def bridges(self) -> Subgraph:
```

Bridges in the full graph.

**bridges\_of****Kind:** Method

```
def bridges_of(self, subgraph: Subgraph) -> Subgraph:
```

Bridges within a subgraph.

**subgraph****Kind:** Parameter

subgraph: Subgraph

**cycle\_basis****Kind:** Method

```
def cycle_basis(self) -> tuple[builtins.list[Cycle], Subgraph]:
```

Cycle basis of the full graph.

#### **cycle\_basis\_of**

**Kind:** Method

```
def cycle_basis_of(self, subgraph: Subgraph) -> tuple[builtins.list[Cycle],  
Subgraph]:
```

Cycle basis of a subgraph.

#### **subgraph**

**Kind:** Parameter

subgraph: Subgraph

#### **all\_spanning\_forests**

**Kind:** Method

```
def all_spanning_forests(self) -> builtins.list[Subgraph]:
```

All spanning forests of the full graph.

#### **all\_spanning\_forests\_of**

**Kind:** Method

```
def all_spanning_forests_of(self, subgraph: Subgraph) -> builtins.list[Subgraph]:
```

All spanning forests of a subgraph.

#### **subgraph**

**Kind:** Parameter

subgraph: Subgraph

#### **combine\_to\_single\_hedgenode**

**Kind:** Method

```
def combine_to_single_hedgenode(self, nodes: typing.Sequence[NodeIndex]) ->  
HedgeNode:
```

Combine nodes into a single hedge node.

#### **nodes**

**Kind:** Parameter

nodes: typing.Sequence[NodeIndex]

#### **all\_cuts**

**Kind:** Method

```
def all_cuts(self, source: HedgeNode, target: HedgeNode) ->  
builtins.list[tuple[Subgraph, OrientedCut, Subgraph]]:
```

All cuts between two hedge nodes.

#### **source**

**Kind:** Parameter

source: HedgeNode

#### **target**

**Kind:** Parameter

target: HedgeNode

### **all\_cuts\_from\_ids**

**Kind:** Method

```
def all_cuts_from_ids(self, source: typing.Sequence[NodeIndex], target:
typing.Sequence[NodeIndex]) -> builtins.list[tuple[Subgraph, OrientedCut, Subgraph]]:
```

All cuts between two sets of node indices.

### **source**

**Kind:** Parameter

source: typing.Sequence[NodeIndex]

### **target**

**Kind:** Parameter

target: typing.Sequence[NodeIndex]

### **contract\_subgraph**

**Kind:** Method

```
def contract_subgraph(self, subgraph: Subgraph, node_data_merge:
typing.Optional[DotVertexData] = None) -> None:
```

Contract a subgraph into a single node, deleting its edges.

### **subgraph**

**Kind:** Parameter

subgraph: Subgraph

### **node\_data\_merge**

**Kind:** Parameter

node\_data\_merge: typing.Optional[DotVertexData] = None

**Default:** None

### **join**

**Kind:** Method

```
def join(self, other: DotGraph, matching_fn: typing.Any, merge_fn: typing.Any) ->
DotGraph:
```

Join two graphs, matching dangling edges via a Python callback.

### **other**

**Kind:** Parameter

other: DotGraph

### **matching\_fn**

**Kind:** Parameter

matching\_fn: typing.Any

### **merge\_fn**

**Kind:** Parameter

merge\_fn: typing.Any

**join\_mut****Kind:** Method

```
def join_mut(self, other: DotGraph, matching_fn: typing.Any, merge_fn: typing.Any) -> None:
```

In-place join, matching dangling edges via a Python callback.

**other****Kind:** Parameter

other: DotGraph

**matching\_fn****Kind:** Parameter

matching\_fn: typing.Any

**merge\_fn****Kind:** Parameter

merge\_fn: typing.Any

**extract****Kind:** Method

```
def extract(self, subgraph: Subgraph, split_edge_fn: typing.Any, internal_data: typing.Any, split_node: typing.Any, owned_node: typing.Any) -> DotGraph:
```

Extract a subgraph with Python callbacks to transform edge/node data.

**subgraph****Kind:** Parameter

subgraph: Subgraph

**split\_edge\_fn****Kind:** Parameter

split\_edge\_fn: typing.Any

**internal\_data****Kind:** Parameter

internal\_data: typing.Any

**split\_node****Kind:** Parameter

split\_node: typing.Any

**owned\_node****Kind:** Parameter

owned\_node: typing.Any

**\_\_getitem\_\_****Kind:** Method

```
def __getitem__(self, key: Hedge) -> DotHedgeData:
```

**key**

**Kind:** Parameter

key: Hedge

**\_\_getitem\_\_**

**Kind:** Method

```
def __getitem__(self, key: NodeIndex) -> DotVertexData:
```

**key**

**Kind:** Parameter

key: NodeIndex

**\_\_getitem\_\_**

**Kind:** Method

```
def __getitem__(self, key: EdgeIndex) -> DotEdgeData:
```

**key**

**Kind:** Parameter

key: EdgeIndex

Exhaustive reference · Source

### **DotGraphBuilder**

Builder for constructing DOT graphs programmatically.

Builder for constructing DOT graphs programmatically.

```
class DotGraphBuilder:
```

**Requires features:** extension-module, abi3-py310

### **Inspect the registered export**

```
from linnet_py import DotGraphBuilder
help(DotGraphBuilder)
```

**\_\_new\_\_**

**Kind:** AssociatedFunction

```
def __new__(cls) -> DotGraphBuilder:
```

Create a new, empty graph builder.

### **add\_node**

**Kind:** Method

```
def add_node(self, data: typing.Optional[DotVertexData] = None) -> NodeIndex:
```

Add a node and return its index.

**data**

**Kind:** Parameter

data: typing.Optional[DotVertexData] = None

**Default:** None

**add\_edge****Kind:** Method

```
def add_edge(self, source: NodeIndex, sink: NodeIndex, data:
typing.Optional[DotEdgeData] = None, orientation: typing.Optional[Orientation] =
None, source_hedge: typing.Optional[DotHedgeData] = None, sink_hedge:
typing.Optional[DotHedgeData] = None) -> None:
```

Add an edge between two nodes.

**source****Kind:** Parameter

source: NodeIndex

**sink****Kind:** Parameter

sink: NodeIndex

**data****Kind:** Parameter

data: typing.Optional[DotEdgeData] = None

**Default:** None**orientation****Kind:** Parameter

orientation: typing.Optional[Orientation] = None

**Default:** None**source\_hedge****Kind:** Parameter

source\_hedge: typing.Optional[DotHedgeData] = None

**Default:** None**sink\_hedge****Kind:** Parameter

sink\_hedge: typing.Optional[DotHedgeData] = None

**Default:** None**add\_external\_edge****Kind:** Method

```
def add_external_edge(self, source: NodeIndex, data: typing.Optional[DotEdgeData] =
None, orientation: typing.Optional[Orientation] = None, flow: typing.Optional[Flow] =
None, hedge: typing.Optional[DotHedgeData] = None) -> None:
```

Add a dangling (external) edge incident to a node.

**source****Kind:** Parameter

source: NodeIndex

**data****Kind:** Parameter

data: typing.Optional[DotEdgeData] = None

**Default:** None**orientation****Kind:** Parameter

orientation: typing.Optional[Orientation] = None

**Default:** None**flow****Kind:** Parameter

flow: typing.Optional[Flow] = None

**Default:** None**hedge****Kind:** Parameter

hedge: typing.Optional[DotHedgeData] = None

**Default:** None**build****Kind:** Method

def build(self) -&gt; DotGraph:

Build the graph and reset the builder.

Exhaustive reference · Source

**DotHedgeData**

DOT hedge (half-edge) data.

DOT hedge (half-edge) data.

class DotHedgeData:

**Requires features:** extension-module, abi3-py310**Inspect the registered export**

```
from linnet_py import DotHedgeData
help(DotHedgeData)
```

**statement****Kind:** Getter

def statement(self) -&gt; typing.Optional[builtins.str]:

Optional statement string.

**id****Kind:** Getter

def id(self) -&gt; typing.Optional[Hedge]:

Optional hedge id.



**port\_label****Kind:** Getter

```
def port_label(self) -> typing.Optional[builtins.str]:  
Optional port label.
```

**compasspt****Kind:** Getter

```
def compasspt(self) -> typing.Optional[builtins.str]:  
Optional compass point as a string.
```

**\_\_new\_\_****Kind:** AssociatedFunction

```
def __new__(cls, statement: typing.Optional[builtins.str] = None, id:  
typing.Optional[builtins.int] = None, port_label: typing.Optional[builtins.str] =  
None, compasspt: typing.Optional[builtins.str] = None) -> DotHedgeData:  
Create hedge data from fields.
```

**statement****Kind:** Parameter

```
statement: typing.Optional[builtins.str] = None
```

**Default:** None**id****Kind:** Parameter

```
id: typing.Optional[builtins.int] = None
```

**Default:** None**port\_label****Kind:** Parameter

```
port_label: typing.Optional[builtins.str] = None
```

**Default:** None**compasspt****Kind:** Parameter

```
compasspt: typing.Optional[builtins.str] = None
```

**Default:** None**\_\_repr\_\_****Kind:** Method

```
def __repr__(self) -> builtins.str:  
Debug-style representation.
```

Exhaustive reference · Source

**DotVertexData**

DOT vertex data (name, optional index, and attribute statements).

DOT vertex data (name, optional index, and attribute statements).

```
class DotVertexData:
```

**Requires features:** extension-module, abi3-py310

### Inspect the registered export

```
from linnet_py import DotVertexData
help(DotVertexData)
```

**name**

**Kind:** Getter

```
def name(self) -> typing.Optional[builtins.str]:
Optional vertex name.
```

**index**

**Kind:** Getter

```
def index(self) -> typing.Optional[NodeIndex]:
Optional node index.
```

**statements**

**Kind:** Getter

```
def statements(self) -> NodeStatements:
Attribute statements as a dict-like proxy.
```

**\_\_new\_\_**

**Kind:** AssociatedFunction

```
def __new__(cls, name: typing.Optional[builtins.str] = None, index:
typing.Optional[builtins.int] = None, statements:
typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None) -> DotVertexData:
Create vertex data from fields.
```

**name**

**Kind:** Parameter

```
name: typing.Optional[builtins.str] = None
```

**Default:** None

**index**

**Kind:** Parameter

```
index: typing.Optional[builtins.int] = None
```

**Default:** None

**statements**

**Kind:** Parameter

```
statements: typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None
```

**Default:** None

**add\_statement**

**Kind:** Method

```
def add_statement(self, key: builtins.str, value: builtins.str) -> None:
```

Insert or update an attribute statement.

#### **key**

**Kind:** Parameter

key: builtins.str

#### **value**

**Kind:** Parameter

value: builtins.str

#### **\_\_repr\_\_**

**Kind:** Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

Exhaustive reference · Source

#### **EdgeData**

Edge data with orientation.

Edge data with orientation.

```
class EdgeData:
```

**Requires features:** extension-module, abi3-py310

#### **Inspect the registered export**

```
from linnet_py import EdgeData
help(EdgeData)
```

#### **orientation**

**Kind:** Getter

```
def orientation(self) -> Orientation:
```

Orientation of this edge.

#### **data**

**Kind:** Getter

```
def data(self) -> DotEdgeData:
```

DOT edge data.

#### **\_\_new\_\_**

**Kind:** AssociatedFunction

```
def __new__(cls, orientation: typing.Any, data: typing.Any) -> EdgeData:
```

Create edge data from orientation and DotEdgeData.

#### **orientation**

**Kind:** Parameter

orientation: typing.Any

#### **data**

**Kind:** Parameter

data: typing.Any

**\_\_repr\_\_**

**Kind:** Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

**EdgeIndex**

Edge index identifier.

Edge index identifier.

```
class EdgeIndex:
```

**Requires features:** extension-module, abi3-py310

**Inspect the registered export**

```
from linnet_py import EdgeIndex
help(EdgeIndex)
```

**value**

**Kind:** Getter

```
def value(self) -> builtins.int:
```

Numeric value of this edge index.

**\_\_new\_\_**

**Kind:** AssociatedFunction

```
def __new__(cls, value: builtins.int) -> EdgeIndex:
```

Create an edge index from a zero-based index.

**value**

**Kind:** Parameter

value: builtins.int

**\_\_int\_\_**

**Kind:** Method

```
def __int__(self) -> builtins.int:
```

Convert to int.

**\_\_repr\_\_**

**Kind:** Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

**EdgeStatements**

Dict-like proxy for edge attribute statements.

Dict-like proxy for edge attribute statements.

```
class EdgeStatements:
```

**Requires features:** extension-module, abi3-py310

### Inspect the registered export

```
from linnet_py import EdgeStatements
help(EdgeStatements)
```

**`__getitem__`**

**Kind:** Method

```
def __getitem__(self, key: builtins.str) -> builtins.str:
```

**key**

**Kind:** Parameter

key: builtins.str

**`__setitem__`**

**Kind:** Method

```
def __setitem__(self, key: builtins.str, value: builtins.str) -> None:
```

**key**

**Kind:** Parameter

key: builtins.str

**value**

**Kind:** Parameter

value: builtins.str

**`__delitem__`**

**Kind:** Method

```
def __delitem__(self, key: builtins.str) -> None:
```

**key**

**Kind:** Parameter

key: builtins.str

**`__contains__`**

**Kind:** Method

```
def __contains__(self, key: builtins.str) -> builtins.bool:
```

**key**

**Kind:** Parameter

key: builtins.str

**`__len__`**

**Kind:** Method

```
def __len__(self) -> builtins.int:
```

**`__iter__`**

**Kind:** Method

```
def __iter__(self) -> typing.Iterator[builtins.str]:
```

**get****Kind:** Method

```
def get(self, key: builtins.str, default: typing.Any = None) -> typing.Any:
```

**key****Kind:** Parameter

```
key: builtins.str
```

**default****Kind:** Parameter

```
default: typing.Any = None
```

**Default:** None**keys****Kind:** Method

```
def keys(self) -> builtins.list[builtins.str]:
```

**values****Kind:** Method

```
def values(self) -> builtins.list[builtins.str]:
```

**items****Kind:** Method

```
def items(self) -> builtins.list[tuple[builtins.str, builtins.str]]:
```

**clear****Kind:** Method

```
def clear(self) -> None:
```

**update****Kind:** Method

```
def update(self, other: typing.Mapping[builtins.str, builtins.str]) -> None:
```

**other****Kind:** Parameter

```
other: typing.Mapping[builtins.str, builtins.str]
```

**pop****Kind:** Method

```
def pop(self, key: builtins.str, default: typing.Any = None) -> typing.Any:
```

**key****Kind:** Parameter

```
key: builtins.str
```

**default****Kind:** Parameter

```
default: typing.Any = None
```

**Default:** None

**popitem**

**Kind:** Method

```
def popitem(self) -> tuple[builtins.str, builtins.str]:
```

**setdefault**

**Kind:** Method

```
def setdefault(self, key: builtins.str, default: typing.Optional[builtins.str] = None) -> builtins.str:
```

**key**

**Kind:** Parameter

key: builtins.str

**default**

**Kind:** Parameter

default: typing.Optional[builtins.str] = None

**Default:** None

**\_\_repr\_\_**

**Kind:** Method

```
def __repr__(self) -> builtins.str:
```

Exhaustive reference · Source

**Flow**

Flow direction (Source/Sink).

Flow direction (Source/Sink).

class Flow:

**Requires features:** extension-module, abi3-py310

**Inspect the registered export**

```
from linnet_py import Flow
help(Flow)
```

**source**

**Kind:** AssociatedFunction

```
def source() -> Flow:
```

Flow::Source.

**sink**

**Kind:** AssociatedFunction

```
def sink() -> Flow:
```

Flow::Sink.

**\_\_repr\_\_**

**Kind:** Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

Exhaustive reference · Source

### **GlobalData**

Global graph attributes (graph/node/edge statements).

Global graph attributes (graph/node/edge statements).

```
class GlobalData:
```

**Requires features:** extension-module, abi3-py310

### **Inspect the registered export**

```
from linnet_py import GlobalData
help(GlobalData)
```

**name**

**Kind:** Getter

```
def name(self) -> builtins.str:
```

Graph name.

**name**

**Kind:** Setter

```
def name(self, value: builtins.str) -> None:
```

**value**

**Kind:** Parameter

value: builtins.str

**statements**

**Kind:** Getter

```
def statements(self) -> GlobalStatements:
```

Graph-level statements.

**statements**

**Kind:** Setter

```
def statements(self, value: builtins.dict[builtins.str, builtins.str]) -> None:
```

**value**

**Kind:** Parameter

value: builtins.dict[builtins.str, builtins.str]

**edge\_statements**

**Kind:** Getter

```
def edge_statements(self) -> GlobalStatements:
```

Default edge statements.

**edge\_statements**

**Kind:** Setter



```
def edge_statements(self, value: builtins.dict[builtins.str, builtins.str]) -> None:
```

**value**

**Kind:** Parameter

value: builtins.dict[builtins.str, builtins.str]

**node\_statements**

**Kind:** Getter

```
def node_statements(self) -> GlobalStatements:
```

Default node statements.

**node\_statements**

**Kind:** Setter

```
def node_statements(self, value: builtins.dict[builtins.str, builtins.str]) -> None:
```

**value**

**Kind:** Parameter

value: builtins.dict[builtins.str, builtins.str]

**\_\_new\_\_**

**Kind:** AssociatedFunction

```
def __new__(cls, name: typing.Optional[builtins.str] = None, statements:
typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None, edge_statements:
typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None, node_statements:
typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None) -> GlobalData:
```

Create global data from fields.

**name**

**Kind:** Parameter

name: typing.Optional[builtins.str] = None

**Default:** None

**statements**

**Kind:** Parameter

statements: typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None

**Default:** None

**edge\_statements**

**Kind:** Parameter

edge\_statements: typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None

**Default:** None

**node\_statements**

**Kind:** Parameter

node\_statements: typing.Optional[typing.Mapping[builtins.str, builtins.str]] = None

**Default:** None

**add\_statement****Kind:** Method

```
def add_statement(self, key: builtins.str, value: builtins.str) -> None:
```

Insert or update a graph-level statement.

**key****Kind:** Parameter

key: builtins.str

**value****Kind:** Parameter

value: builtins.str

**add\_edge\_statement****Kind:** Method

```
def add_edge_statement(self, key: builtins.str, value: builtins.str) -> None:
```

Insert or update a default edge statement.

**key****Kind:** Parameter

key: builtins.str

**value****Kind:** Parameter

value: builtins.str

**add\_node\_statement****Kind:** Method

```
def add_node_statement(self, key: builtins.str, value: builtins.str) -> None:
```

Insert or update a default node statement.

**key****Kind:** Parameter

key: builtins.str

**value****Kind:** Parameter

value: builtins.str

**\_\_repr\_\_****Kind:** Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

**GlobalStatements**

Dict-like proxy for global statements.

Dict-like proxy for global statements.

class GlobalStatements:

**Requires features:** extension-module, abi3-py310

### Inspect the registered export

```
from linnet_py import GlobalStatements
help(GlobalStatements)
```

**\_\_getitem\_\_**

**Kind:** Method

```
def __getitem__(self, key: builtins.str) -> builtins.str:
```

**key**

**Kind:** Parameter

key: builtins.str

**\_\_setitem\_\_**

**Kind:** Method

```
def __setitem__(self, key: builtins.str, value: builtins.str) -> None:
```

**key**

**Kind:** Parameter

key: builtins.str

**value**

**Kind:** Parameter

value: builtins.str

**\_\_delitem\_\_**

**Kind:** Method

```
def __delitem__(self, key: builtins.str) -> None:
```

**key**

**Kind:** Parameter

key: builtins.str

**\_\_contains\_\_**

**Kind:** Method

```
def __contains__(self, key: builtins.str) -> builtins.bool:
```

**key**

**Kind:** Parameter

key: builtins.str

**\_\_len\_\_**

**Kind:** Method

```
def __len__(self) -> builtins.int:
```

**`__iter__`**

**Kind:** Method

```
def __iter__(self) -> typing.Iterator[builtins.str]:
```

**get**

**Kind:** Method

```
def get(self, key: builtins.str, default: typing.Any = None) -> typing.Any:
```

**key**

**Kind:** Parameter

key: builtins.str

**default**

**Kind:** Parameter

default: typing.Any = None

**Default:** None

**keys**

**Kind:** Method

```
def keys(self) -> builtins.list[builtins.str]:
```

**values**

**Kind:** Method

```
def values(self) -> builtins.list[builtins.str]:
```

**items**

**Kind:** Method

```
def items(self) -> builtins.list[tuple[builtins.str, builtins.str]]:
```

**clear**

**Kind:** Method

```
def clear(self) -> None:
```

**update**

**Kind:** Method

```
def update(self, other: typing.Mapping[builtins.str, builtins.str]) -> None:
```

**other**

**Kind:** Parameter

other: typing.Mapping[builtins.str, builtins.str]

**pop**

**Kind:** Method

```
def pop(self, key: builtins.str, default: typing.Any = None) -> typing.Any:
```

**key**

**Kind:** Parameter

key: builtins.str

**default****Kind:** Parameter

default: typing.Any = None

**Default:** None**popitem****Kind:** Method

def popitem(self) -&gt; tuple[builtins.str, builtins.str]:

**setdefault****Kind:** Method

def setdefault(self, key: builtins.str, default: typing.Optional[builtins.str] = None) -&gt; builtins.str:

**key****Kind:** Parameter

key: builtins.str

**default****Kind:** Parameter

default: typing.Optional[builtins.str] = None

**Default:** None**\_\_repr\_\_****Kind:** Method

def \_\_repr\_\_(self) -&gt; builtins.str:

Exhaustive reference · Source

**Hedge**

Half-edge identifier.

Half-edge identifier.

class Hedge:

**Requires features:** extension-module, abi3-py310**Inspect the registered export**

```
from linnet_py import Hedge
help(Hedge)
```

**value****Kind:** Getter

def value(self) -&gt; builtins.int:

Numeric value of this hedge.

**\_\_new\_\_****Kind:** AssociatedFunction

def \_\_new\_\_(cls, value: builtins.int) -&gt; Hedge:

Create a hedge from a zero-based index.

**value**

**Kind:** Parameter

value: builtins.int

**\_\_int\_\_**

**Kind:** Method

def \_\_int\_\_(self) -> builtins.int:

Convert to int.

**\_\_repr\_\_**

**Kind:** Method

def \_\_repr\_\_(self) -> builtins.str:

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

**HedgeNode**

Hedge node with internal graph and hairs.

Hedge node with internal graph and hairs.

class HedgeNode:

**Requires features:** extension-module, abi3-py310

**Inspect the registered export**

```
from linnet_py import HedgeNode
help(HedgeNode)
```

**internal\_graph**

**Kind:** Getter

def internal\_graph(self) -> Subgraph:

Internal subgraph.

**hairs**

**Kind:** Getter

def hairs(self) -> Subgraph:

Hair subgraph.

**\_\_new\_\_**

**Kind:** AssociatedFunction

def \_\_new\_\_(cls, internal\_graph: Subgraph, hairs: Subgraph) -> HedgeNode:

Create a hedge node from internal graph and hairs subgraphs.

**internal\_graph**

**Kind:** Parameter

internal\_graph: Subgraph

## **hairs**

**Kind:** Parameter

hairs: Subgraph

## **\_\_repr\_\_**

**Kind:** Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

## **HedgePair**

Hedge pairing (paired, unpaired, split).

Hedge pairing (paired, unpaired, split).

```
class HedgePair:
```

**Requires features:** extension-module, abi3-py310

## **Inspect the registered export**

```
from linnet_py import HedgePair
help(HedgePair)
```

## **kind**

**Kind:** Getter

```
def kind(self) -> builtins.str:
```

Kind of hedge pair: \"paired\", \"unpaired\", or \"split\".

## **source**

**Kind:** Getter

```
def source(self) -> typing.Optional[Hedge]:
```

Source hedge (paired/split).

## **sink**

**Kind:** Getter

```
def sink(self) -> typing.Optional[Hedge]:
```

Sink hedge (paired/split).

## **hedge**

**Kind:** Getter

```
def hedge(self) -> typing.Optional[Hedge]:
```

Hedge (unpaired only).

## **flow**

**Kind:** Getter

```
def flow(self) -> typing.Optional[Flow]:
```

Flow for unpaired hedge.

## **split**

**Kind:** Getter

```
def split(self) -> typing.Optional[Flow]:
```

Which side is split for split edges.

**\_\_repr\_\_**

**Kind:** Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

## NodeIndex

Node index identifier.

Node index identifier.

```
class NodeIndex:
```

**Requires features:** extension-module, abi3-py310

## Inspect the registered export

```
from linnet_py import NodeIndex
help(NodeIndex)
```

**value**

**Kind:** Getter

```
def value(self) -> builtins.int:
```

Numeric value of this node index.

**\_\_new\_\_**

**Kind:** AssociatedFunction

```
def __new__(cls, value: builtins.int) -> NodeIndex:
```

Create a node index from a zero-based index.

**value**

**Kind:** Parameter

value: builtins.int

**\_\_int\_\_**

**Kind:** Method

```
def __int__(self) -> builtins.int:
```

Convert to int.

**\_\_repr\_\_**

**Kind:** Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

## NodeStatements

Dict-like proxy for node attribute statements.

Dict-like proxy for node attribute statements.



```
class NodeStatements:
```

**Requires features:** extension-module, abi3-py310

### Inspect the registered export

```
from linnet_py import NodeStatements  
help(NodeStatements)
```

**\_\_getitem\_\_**

**Kind:** Method

```
def __getitem__(self, key: builtins.str) -> builtins.str:
```

**key**

**Kind:** Parameter

key: builtins.str

**\_\_setitem\_\_**

**Kind:** Method

```
def __setitem__(self, key: builtins.str, value: builtins.str) -> None:
```

**key**

**Kind:** Parameter

key: builtins.str

**value**

**Kind:** Parameter

value: builtins.str

**\_\_delitem\_\_**

**Kind:** Method

```
def __delitem__(self, key: builtins.str) -> None:
```

**key**

**Kind:** Parameter

key: builtins.str

**\_\_contains\_\_**

**Kind:** Method

```
def __contains__(self, key: builtins.str) -> builtins.bool:
```

**key**

**Kind:** Parameter

key: builtins.str

**\_\_len\_\_**

**Kind:** Method

```
def __len__(self) -> builtins.int:
```

**\_\_iter\_\_**

**Kind:** Method

```
def __iter__(self) -> typing.Iterator[builtins.str]:
```

**get**

**Kind:** Method

```
def get(self, key: builtins.str, default: typing.Any = None) -> typing.Any:
```

**key**

**Kind:** Parameter

key: builtins.str

**default**

**Kind:** Parameter

default: typing.Any = None

**Default:** None

**keys**

**Kind:** Method

```
def keys(self) -> builtins.list[builtins.str]:
```

**values**

**Kind:** Method

```
def values(self) -> builtins.list[builtins.str]:
```

**items**

**Kind:** Method

```
def items(self) -> builtins.list[tuple[builtins.str, builtins.str]]:
```

**clear**

**Kind:** Method

```
def clear(self) -> None:
```

**update**

**Kind:** Method

```
def update(self, other: typing.Mapping[builtins.str, builtins.str]) -> None:
```

**other**

**Kind:** Parameter

other: typing.Mapping[builtins.str, builtins.str]

**pop**

**Kind:** Method

```
def pop(self, key: builtins.str, default: typing.Any = None) -> typing.Any:
```

**key**

**Kind:** Parameter

key: builtins.str

**default****Kind:** Parameter

default: typing.Any = None

**Default:** None**popitem****Kind:** Method

def popitem(self) -&gt; tuple[builtins.str, builtins.str]:

**setdefault****Kind:** Method

def setdefault(self, key: builtins.str, default: typing.Optional[builtins.str] = None) -&gt; builtins.str:

**key****Kind:** Parameter

key: builtins.str

**default****Kind:** Parameter

default: typing.Optional[builtins.str] = None

**Default:** None**\_\_repr\_\_****Kind:** Method

def \_\_repr\_\_(self) -&gt; builtins.str:

Exhaustive reference · Source

**Orientation**

Edge orientation (Default/Reversed/Undirected).

Edge orientation (Default/Reversed/Undirected).

class Orientation:

**Requires features:** extension-module, abi3-py310**Inspect the registered export**

```
from linnet_py import Orientation
help(Orientation)
```

**default****Kind:** AssociatedFunction

def default() -&gt; Orientation:

Orientation::Default.

**reversed****Kind:** AssociatedFunction

def reversed() -&gt; Orientation:

Orientation::Reversed.

### **undirected**

**Kind:** AssociatedFunction

```
def undirected() -> Orientation:
```

Orientation::Undirected.

### **\_\_repr\_\_**

**Kind:** Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

Exhaustive reference · Source

### **OrientedCut**

Oriented cut represented by left/right subgraphs.

Oriented cut represented by left/right subgraphs.

```
class OrientedCut:
```

**Requires features:** extension-module, abi3-py310

### **Inspect the registered export**

```
from linnet_py import OrientedCut
```

```
help(OrientedCut)
```

### **left**

**Kind:** Getter

```
def left(self) -> Subgraph:
```

Left side of the cut.

### **right**

**Kind:** Getter

```
def right(self) -> Subgraph:
```

Right side of the cut.

### **\_\_repr\_\_**

**Kind:** Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

Exhaustive reference · Source

### **Subgraph**

Subgraph represented as a bitset of hedges.

Subgraph represented as a bitset of hedges.

```
class Subgraph:
```

**Requires features:** extension-module, abi3-py310

### **Inspect the registered export**

```
from linnet_py import Subgraph
help(Subgraph)
```

#### **empty**

**Kind:** AssociatedFunction

```
def empty(cls, size: builtins.int) -> Subgraph:
```

Create an empty subgraph of the given size (number of hedges).

#### **size**

**Kind:** Parameter

size: builtins.int

#### **full**

**Kind:** AssociatedFunction

```
def full(cls, size: builtins.int) -> Subgraph:
```

Create a full subgraph including all hedges.

#### **size**

**Kind:** Parameter

size: builtins.int

#### **from\_hedges**

**Kind:** AssociatedFunction

```
def from_hedges(cls, size: builtins.int, hedges: typing.Sequence[Hedge]) -> Subgraph:
```

Create a subgraph from a list of hedges.

#### **size**

**Kind:** Parameter

size: builtins.int

#### **hedges**

**Kind:** Parameter

hedges: typing.Sequence[Hedge]

#### **to\_hedges**

**Kind:** Method

```
def to_hedges(self) -> builtins.list[Hedge]:
```

List included hedges.

#### **size**

**Kind:** Method

```
def size(self) -> builtins.int:
```

Total number of hedges in the parent graph.

#### **n\_included**

**Kind:** Method

```
def n_included(self) -> builtins.int:
```

Number of included hedges.

### **includes**

**Kind:** Method

```
def includes(self, hedge: typing.Any) -> builtins.bool:
```

Whether a hedge is included.

### **hedge**

**Kind:** Parameter

hedge: typing.Any

### **union**

**Kind:** Method

```
def union(self, other: Subgraph) -> Subgraph:
```

Union with another subgraph.

### **other**

**Kind:** Parameter

other: Subgraph

### **intersection**

**Kind:** Method

```
def intersection(self, other: Subgraph) -> Subgraph:
```

Intersection with another subgraph.

### **other**

**Kind:** Parameter

other: Subgraph

### **sym\_diff**

**Kind:** Method

```
def sym_diff(self, other: Subgraph) -> Subgraph:
```

Symmetric difference with another subgraph.

### **other**

**Kind:** Parameter

other: Subgraph

### **subtract**

**Kind:** Method

```
def subtract(self, other: Subgraph) -> Subgraph:
```

Subtract another subgraph.

### **other**

**Kind:** Parameter

other: Subgraph

**\_\_repr\_\_**

**Kind:** Method

```
def __repr__(self) -> builtins.str:
```

Debug-style representation.

[Exhaustive reference](#) · [Source](#)

**TraversalTree**

Traversal tree produced by DFS/BFS.

Traversal tree produced by DFS/BFS.

```
class TraversalTree:
```

**Requires features:** extension-module, abi3-py310

**Inspect the registered export**

```
from linnet_py import TraversalTree
help(TraversalTree)
```

**tree\_subgraph**

**Kind:** Method

```
def tree_subgraph(self) -> Subgraph:
```

Subgraph corresponding to the tree edges.

**node\_order**

**Kind:** Method

```
def node_order(self) -> builtins.list[NodeIndex]:
```

Node order from traversal.

**covers**

**Kind:** Method

```
def covers(self, subgraph: Subgraph) -> Subgraph:
```

Covers a subgraph with the traversal tree.

**subgraph**

**Kind:** Parameter

subgraph: Subgraph

**iter\_hedges**

**Kind:** Method

```
def iter_hedges(self) -> builtins.list[tuple[Hedge, builtins.str,
typing.Optional[Hedge]]]:
```

Iterate hedges as (hedge, kind, root\_hedge).

[Exhaustive reference](#) · [Source](#)

# Canonical package changelog

The following release history is imported as typed Markdown from crates/linnet/CHANGELOG.md.

## Changelog

All notable changes to this project will be documented in this file.

The format is based on Keep a Changelog, and this project adheres to Semantic Versioning.

### [Unreleased]

#### 0.17.0 - 2026-01-28

##### Other

- force-directed layout
- make the powerset iterator properly size the subset
- upgrade to all spanning *\*forests\** generation
- Refactor PowersetIterator to support generic ID type
- double ended iterator for subsets
- remove unused crates

#### 0.16.1 - 2026-01-13

##### Other

- Quote string values in graph and parser outputs
- revert edge, node and hedge index order to be direct (not based on sort)

#### 0.16.0 - 2026-01-08

##### Other

- Add methods for identifying bridges and non-bridges in subgraphs
- Make `trace_unfold` implementable outside of linnet

#### 0.15.1 - 2026-01-05

##### Fixed

- fixed ordering problem in merge function and make ordering based on sorting not on direct map

##### Other

- update quote stripping
- Add agent context files and update project tooling
- Add graph algorithms module with topological sort and transitive ops

#### 0.15.0 - 2025-12-08

##### Other

- Refactor Figment handling and improve data merging logic
- *\*(clinnet)\** release v0.1.8

#### 0.14.1 - 2025-12-03

##### Other

- Add parallel processing and directional constraints
- update test



- only swap one edge when connecting identities

**0.14.0 - 2025-11-28**

**Other**

- change output name and make it short, and add directional groups

**0.13.3 - 2025-11-24**

**Other**

- update symbolica

**0.13.2 - 2025-11-12**

**Other**

- update to symbolica 0.20
- better templates
- clinnet for nested dots to pdf
- Add Figment integration for configuration handling

**0.13.1 - 2025-11-05**

**Other**

- remove prints

**0.12.0 - 2025-08-18**

**Other**

- all tests pass
- clippy fixes
- write global data outside of graph[] scope
- Add ID=ID statements to global data when parsing
- newline after nodes
- dot serialize with name
- write graph name
- correctly parse orientations
- make iter\_edges iterate with edge\_id order not involution order

**0.11.0 - 2025-07-29**

**Other**

- serialize global data

**0.10.0 - 2025-07-29**

**Other**

- Add check graph function and extracting by nodes
- Add is\_empty to Swap trait and update implementations

**0.9.1 - 2025-07-28**

**Other**

- impl asref for newtypes

## **0.9.0 - 2025-07-27**

### **Other**

- Allow specifying a subgraph in the DOT format.
- Rewrite DOT output to include node, hedge and edge indices and fix bidirectional serialization
- DotGraph has concrete data. Parser now supports global edge and vertex data. Can set edge, vertex and hedge index
- add multi-graph parsing and global graph data

## **0.8.0 - 2025-07-14**

### **Other**

- clippy fix
- Rename HedgeVec to EdgeVec and introduce HedgeVec
- Add H parameter to HedgeGraph in iter\_edges and orientation
- add H to SignedCycle
- rename as\_ref to to\_ref
- Add hedge\_data to hedge\_graph
- add perm by key
- update symbolica
- Fix tree traversal and modify ForestNodeStore API to use Option

## **0.7.1 - 2025-05-24**

### **Other**

- Fix crown computation by inverting subgraph inclusion check and add inclusion for HedgePair

## **0.7.0 - 2025-05-23**

### **Other**

- unify iterator signature and remove redundant and misleading functions
- Add zenodo doi

## **0.6.3 - 2025-05-23**

### **Other**

- improve README

## **0.6.2 - 2025-05-22**

### **Other**

- remove useless check and set
- Correctly forget identification history and adds subgraph contraction

## **0.6.1 - 2025-05-22**

### **Other**

- Merge pull request #11 from alpha00p/jules\_wip\_3253120278989361716

## **0.6.0 - 2025-05-22**

### **Fixed**

- fix all tests and update symbolica

- fixes forget\_identification\_history, If you delete a subgraph you need

#### **Other**

- Add crown functions and implement spanning tree enumeration algo based on graph contraction
- PartialEq and Eq for hedgevec
- from\_raw for hedgevec

#### **0.5.2 - 2025-05-09**

##### **Fixed**

- fix involution extraction

##### **Other**

- Add Permutation iterators, new graph extract test

#### **0.5.1 - 2025-05-05**

##### **Other**

- Clean up debug prints and add cycle test

#### **0.5.0 - 2025-05-05**

##### **Fixed**

- fixes all around

##### **Other**

- boundary->crown
- hairs->boundary

#### **0.4.0 - 2025-04-30**

##### **Added**

- feature flag bincode and serde

#### **0.3.0 - 2025-04-30**

##### **Fixed**

- fixed extraction for childvec forest, all tests pass and fix warnings

##### **Other**

- adds extract to SmartHedgeVec and involution
- maps now take a FnMut
- derive more (Decode and Encode)

#### **0.2.3 - 2025-04-08**

##### **Other**

- unpin serde

#### **0.2.2 - 2025-04-01**

##### **Fixed**

- fix test snapshots

**Other**

- raw image

**0.2.1 - 2025-04-01****Other**

- Update README.md, logo is a github asset
- put crates badge

**0.2.0 - 2025-04-01****Fixed**

- fix cetz orientation

**Other**

- Create release-plz.yml

## Version information

Save these identifiers with published results and include them in issue reports so that collaborators can reproduce the same software environment.

- **Documentation channel:** latest
- **Documentation route:** /products/linnet/latest/
- **Source revision:** 0e4439b10bd784f4488a77a024032e3424082631

## Package versions and enabled features

- gammaloop/gammalooprs — 0.3.4 (no optional features)
- gammaloop/gammaloop-api — 0.1.0 (features: cli)
- gammaloop/gammaloop — 0.3.4 (features: python\_api, python\_abi, python\_stubgen)
- linnet/linnet — 0.17.0 (features: nodestore-vec, serde, bincode, drawing, symbolica)
- linnet/linnet-py — 0.1.0 (features: extension-module, abi3-py310)
- spenso/spenso — 0.6.0 (features: shadowing)
- spenso/spenso-macros — 0.3.2 (features: shadowing)
- spenso/spenso-hep-lib — 0.1.7 (no optional features)
- spenso/spynso3 — 0.1.2 (features: python\_stubgen)
- idenso/idenso — 0.3.0 (features: bincode, reference-cases)
- idenso/idenso — 0.3.0 (features: python, python\_stubgen)
- vakint/vakint — 0.1.2 (no optional features)
- vakint/vakint — 0.1.2 (features: symbolica\_community\_module, python\_stubgen)